

Athul Menon P

+91 7025192439 | Thrissur, Kerala, India | athul.menon10@gmail.com
Portfolio | LinkedIn | GitHub

Professional Summary

Software Engineer with experience in full-stack development, backend systems, and AI-driven applications. Skilled in Python, JavaScript, REST APIs, cloud-native development, and agentic AI frameworks. Experienced in enterprise backend modernization, system analysis, and scalable application design.

Skills

- **Languages:** Python, Java, JavaScript
- **Backend:** Node.js, Express.js, REST APIs
- **Frontend:** React.js
- **AI/ML:** OpenCV, TensorFlow, PyTorch, Agentic AI frameworks, LLM Orchestration, Prompt Engineering
- **Databases:** MongoDB, MySQL, PostgreSQL, IBM DB2
- **Cloud & DevOps:** Google Cloud Platform, CI/CD
- **Tools:** Git, GitHub
- **Soft Skills:** Communication, Teamwork, Problem Solving, Attention to Detail, Quick Learner

Education

Bachelor of Technology (B.Tech)

Vidya Academy of Science and Technology, 2020 – 2024

Computer Science and Engineering

Experience

Software Developer - I

December 2024 – Present

UST

- Designed and delivered a high-throughput retail coupon processing system, modernizing a legacy platform and supporting **millions of retail transactions per month**
- Implemented core backend business logic to process **10K+ coupon validations per hour**, ensuring strong transactional integrity, data consistency, and accuracy under peak load
- Built modular, reusable backend components using a procedure- and service-based architecture, following enterprise-grade modularization and interface standards
- Partnered with **business stakeholders** to translate requirements into detailed system designs and conducted technical walkthroughs for new and migrated workflows
- Optimized backend processing and database access patterns, reducing peak-time latency by **30%** and significantly improving system reliability

Projects

Smart Scholar: Agentic Deep Research Assistant [\[GitHub\]](#)

- Built an **agentic deep-research system** using the **OpenAI Agents SDK**, coordinating Planner, Search, Writer, and Email agents to automate multi-step research workflows.
- Applied **asynchronous orchestration** for concurrent web search and analysis, reducing end-to-end research time by **60%**.
- Integrated live web search and automated report generation with structured, long-form outputs, delivering insights via a **web interface and email automation**.

Prism - Real-Time Chat Application [\[GitHub\]](#)

- Developed a **full-stack real-time chat application** supporting concurrent users with bidirectional communication using the MERN stack.
- Implemented **real-time messaging and presence features** (online status, session sync) using **Socket.io**, ensuring low-latency updates across clients.
- Built secure authentication and authorization workflows using **JWT**, enforcing protected routes and controlled data access.

Career_Conversation: AI-Powered Interactive Resume Agent [\[Live Demo\]](#)

- Built an **AI-powered interactive resume agent** enabling real-time, conversational exploration of career history and projects using a large language model.
- Integrated **Gemini LLM** via OpenAI-compatible APIs to create a **model-agnostic, modular architecture**, allowing easy LLM replacement and extension.
- Deployed a **24/7 publicly accessible AI application** on Hugging Face Spaces using Gradio, with automated user alerting for contact requests.

DragonSense: AI-Driven Natural Language Query System

- Built an **AI-driven web application** enabling users to query relational databases using human voice, translating inputs into SQL queries.
- Developed backend services using **Flask** and **MySQL**, optimizing query execution and database interactions.
- Created a responsive and intuitive frontend using **HTML, Bootstrap, and jQuery** to ensure smooth and accessible user experience.

Certifications

- Cloud Digital Leader (GCP) – Google
- The Complete Full-Stack Web Developer Bootcamp – Udemy
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional – Oracle
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle
- The AI Engineer Agentic Track: The Complete Agent & MCP Course – Udemy